

Jad El Harake

jne17@mail.aub.edu | (+961) 70 005 477 | Beirut, Lebanon

EDUCATION

American University of Beirut (AUB) | *Beirut, Lebanon*

Expected Graduation: Jun. 2027

- B.E. in Computer and Communication Engineering
- Member of the ACM club, and IEEE club
- Relevant Courses: EECE 490 (Intro. to Machine Learning), EECE 503N (MLOps-LLMOps), EECE 693 (Neural Network), EECE 433 (Database Systems), EECE 230 (Introduction to Programming), EECE 330 (Data Structure and Algorithms), EECE 332 (Object Oriented and Effective Java Programming),

Grand Lycée Franco Libanais de Beirut | *Beirut, Lebanon*

Sep. 2008 – Jun. 2022

- Activities and societies: Sport (Football Varsity Team player), Member of the Senior Committee

PROFESSIONAL EXPERIENCE

Invigo, DBMS Engineer Intern | *Dekwaneh, Lebanon*

Jul. 2025 – Sep. 2025

- Optimized a national telecom PostgreSQL device database (millions of records) with SQL rules to validate VoLTE/VoWiFi fields (Linux VM).
- Built Python automation for IMEI/TAC parsing and compatibility + statistical reporting at scale.
- Applied GSMA (IR.92/IR.94) and 3GPP standards to define IMS/SIP/HD Voice logic; added a ChatGPT API checker achieving 92% validation consistency.

AUB, Research Assistant (Advisor: Prof. Mariette Awad) | *Beirut, Lebanon*

Aug. 2025 – Dec. 2025

- Mapped Errorless Learning + Reminiscence Therapy into concrete AI workflows (ID + tagging + LLM cues/storytelling + adaptive/emotion-aware behavior).
- Co-designed a microservice backend and finalized an ERD (30+ entities) linking users/clinical data/media and AI interactions.
- Built early prototypes: YOLO-style face detection, embedding + cosine-similarity verification, Bi-LSTM text progress scoring, and FastAPI clinical scheduling/availability + family media flows.

AUB, Undergraduate Research | *Beirut, Lebanon*

Mar. 2025 – Dec. 2025

- Researched 10+ key image segmentation papers, conducting a systematic analysis of state-of-the-art architectures as part of an undergraduate research initiative at AUB.

PROJECTS

Library Database Management System | *EECE 433 (Database Systems)*

- Designed a multi-branch library database (books/copies, members, staff, loans, reservations, payments/fines).
- Built ERD, normalized schema and implemented SQL DDL with strong constraints (PK/FK, CHECK, referential actions).
- Automated workflows with procedures + triggers (loan/reservation limits, one active loan per copy, auto fine calculation).

Online Multiplayer Racing Game (Python, PyQt5) | *EECE 350 (Networking)*

- Developed a real-time two-player racing game using Python socket programming with hybrid client-server/peer-to-peer architecture.
- Built the full GUI using PyQt5, featuring login/registration, user chat, matchmaking, live race visuals, and health-based collision logic.
- Added multimedia features including background music and sound effects using PyQt5's multimedia modules.
- Integrated SQLite for persistent user accounts and win tracking on the server side.

VideoGraph AI Scene Graph | *EECE 490 (Machine Learning)*

- Sampled video frames and used a VLLM to generate captions + structured scene-graph JSON.
- Used YOLO detections (boxes/crops) to guide the VLLM toward key objects and interactions.
- Parsed the JSON into an interactive node-edge graph, with entity merging + stable IDs across frames.

EXTRACURRICULAR ACTIVITIES

ACM Cybersecurity, Participant | *Beirut, Lebanon*

Jan. 2025 – Present

- Participated in Lebanon's First National University CTF organized by ACM CYBER and Semicolon Academy

SKILLS & INTERESTS

Languages: Arabic (Native) | English (Fluent) | French (Fluent)

Computer Skills: Python, C++, Java, Jira, MySQL, PostgreSQL, Git/GitHub, Spring Boot, Docker, MLflow, Prometheus, Grafana

Interests: AI & Technology, Fitness & Performance Science, Soccer, Basketball, Reading